

Accountability Audit Packet (Template)

Decision Provenance & Outcome Attribution for Algorithmic Trade Decisions

Prepared: January 25, 2026

This document is a template packet designed to support trade-governance and accountability audits (e.g., review of decision traceability, controls, and outcome attribution). It is not legal advice and does not represent any regulator's required format. Populate it with your actual artifacts generated by the pipeline described herein.

1. Scope and Objectives

Objective: Provide a reproducible, end-to-end trace from (a) inputs and constraints, (b) model outputs and learned decision factors, through (c) executed actions, and (d) measured outcomes, with per-factor win/loss attribution scoped to ENTRY, EXIT, and SIZING decisions.

2. Artifact Inventory (Evidence)

Artifact	Purpose	Format	Minimum Contents
decision_trace.jsonl	Per-decision event log (ENTRY/EXIT/SIZE)	JSON Lines	Schema v1, hashes, timestamps, factors, outcomes
decision_factor_attribution.csv	Aggregated attribution metrics per factor	CSV	Scope, wins/losses, expectancy, counts, provenance
inputs_schema.json	Baked-in input schema and transforms	JSON	Feature names, transforms, timeframes, versioning
entry_rules.md / exit_rules.md / sizing_rules.md	Hard-coded surrogate conditions	Markdown	Rule list + surrogate fidelity metrics
fuzzy_system.json	Fuzzy membership/rules (if used)	JSON	Membership functions, rule base, defuzz method
model_card.md	Model purpose and limitations	Markdown	Training data window, constraints, known failure modes
run_manifest.json	Reproducibility manifest	JSON	Git commit, environment, dependency lock, dataset hashes

3. Control Objectives and Audit Questions

Control Objective	Audit Question	Evidence
Traceability	Can every action be traced to inputs, rules, and model outputs at time of decision?	decision_trace.jsonl, run_manifest.json
Explainability	Can major drivers of each decision be identified and summarized?	decision_factor_attribution_*.csv, *_rules.md
Reproducibility	Can the run be reproduced from code + data hashes and env?	run_manifest.json, lockfiles
Risk Governance	Are hard risk limits enforced and logged (gates, blocks, overrides)?	decision_trace.jsonl, risk_config.json
Outcome Attribution	Can factor-level performance be quantified separately for ENTRY/EXIT/SIZE?	decision_factor_attribution.csv

4. Decision Taxonomy and Win/Loss Definitions (Summary)

Decisions are scoped into three disjoint categories: ENTRY (open/enter position), EXIT (close/exit or materially reduce exposure), and SIZING (position size multiplier / contracts). Win/Loss must be computed separately per scope to avoid conflating entry quality with exit timing or size risk.

Scope	Win Definition (canonical)	Loss Definition (canonical)
ENTRY	First-passage: reach $+\alpha R$ before $-\beta R$ within evaluation window.	First-passage: reach $-\beta R$ before $+\alpha R$ (or expire without reaching $+\alpha R$).
EXIT	Utility of exiting now > utility of holding Δ bars (risk-adjusted).	Utility of exiting now < utility of holding Δ bars (risk-adjusted).
SIZING	Utility under chosen size m > utility under baseline size m_0 .	Utility under chosen size m < utility under baseline size m_0 .

5. Sign-off and Exception Log (Template)

Record approvals and any exceptions (manual overrides, data gaps, forced exits). Attach supporting evidence and link each exception to `decision_id(s)` in `decision_trace.jsonl`.

Role	Name	Date	Signature / Attestation
Model Owner			
Risk Owner			
DevOps / Platform			
Compliance Reviewer (optional)			